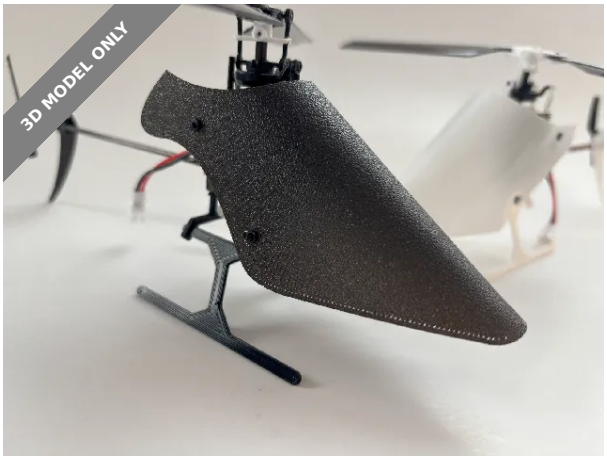

Blade mSRX RC Heli Parts

Fadi

VIEW IN BROWSER

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Summary

Almost complete set of parts for the Blade mSRX RC Helicopter!

Hobby & Makers > RC & Robotics

Tags: rc rchelicopter rcparts rcheli msrx

Updates:

- Dec 22, 2024
  - Added image showing parts in assembled state.
  - Fixed stop tabs on skids (they were too far apart).
  - Fixed swashplate and main rotor hub links (inconsistent ball offset)
  - Updated notes

Decided to model my Blade mSRX RC helicopter after getting tired of having to purchase replacement parts (yes I've crashed it countless times!)

This obviously only replaces the plastic parts (with the exception of the gears). You'll still need your original (or replacement) hardware such as bearings, o-rings, feathering shaft, screws, motors.

Hope this will be useful for some people that want to dust-off their old mSRX heli ;-)

Printing

- Layer height: 0.2mm layer height should do the trick for most of these parts. Consider using 0.1mm layer height for linkages, swashplate, and rotor grip and hub.
- Supports: Most parts can be printed as-is without supports. Exceptions:
    - canopy mount: needs some support for the overhang part that slides over the swashplate pin guide on the main body.
    - rotor grip: print supports inside the space where the rotor blade would slide in. The supports will have to be removed with a box cutter, hobby knife, etc.
    - swashplate (inner and outer): needs supports under the ball links
- I've included 2 sets of linkages. Ones with 2.2mm diameter and ones for 2.3mm, use whichever works better with the finished ball links (see ball link post-processing below).

Post-Processing / Assembly Notes

- Balls on blade grip and swashplate
    - Use a nail file to smooth the balls. I almost always also coat it with a very thin layer of crazy glue and re-file as needed to get as close to a perfect sphere as possible.
- Ball links
    - These are not symmetrical, the side facing down in the model is the one that should be snapping onto the ball.
- Canopy top mount
    - Use a bit of hot glue to hold the mount in place on top of the main body. Could use crazy glue but hot glue makes it easier to replace in case it breaks following a crash :-)
- Swashplate
    1. Insert bearing into (outer) swashplate with a tiny dab of crazy glue on the wall of the swashplate body (careful not to get it into the bearing!)
    2. Push the swashplate ball into the printed swashplate ring.
    3. Slide the ball/ring into the swashplate inner piece
    4. Push the swashplate inner piece into the bearing of step 1
    5. NOTE: You can use “swashplate_inner_alt.stl” instead which eliminates the need to use the ball bearing from the stock swashplate. However, there will be some play which will affect flight performance.
- Main rotor hub
    - Use original o-rings, washers, feathering shaft.
    - Will probably need to use a rotary tool with a small bit to adjust the hole for the feathering shaft.

- Blade grip
    - Remove supports with a precision knife, file the gap where the rotors blade will be inserted using a nail file to made it smooth.
    - Screw holes may need to be slightly widened. I use a heated needle pin if needed.
- Canopy
    - Optional - Bend by hand to obtain desired curvature and heat with a hair dryer. This is optional as the rubber grommets will hold it in place regardless, but I find that it looks nicer when shaped/heated.

Miscellaneous

- The 3D-printed rotors work, but not as effectively as the originals (I'm still working on getting a similar performance.). For instance, with 300mAh batteries, I get 7m of flight time with the original rotors but only 5m30s on the 3D printed ones.
- I've included an alternate front skid mount that can hold larger batteries (I'm using 300mAh LIHV batteries).
- I've used 2mm carbon rods from AliExpress as tail rod replacement

Model files

body.stl

main_rotor_hub.stl

blade_grip.stl

swashplate_outer.stl

swashplate_inner.stl

swashplate_ring.stl

main_shaft_collar.stl

tail_rotor_mount.stl

canopy_mount.stl

skid_mount_front.stl

skid_mount_rear.stl

canopy.stl

main_rotor_blade.stl

tail_rotor_blade.stl

tail_fin.stl

skid_left.stl

skid_right.stl

swashplate_link_22.stl

swashplate_link_23.stl

main_rotor_hub_link_22.stl

main_rotor_hub_link_23.stl

swashplate_pin_guide_replacement.stl

skid_mount_front_alt.stl

swashplate_inner_alt.stl

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